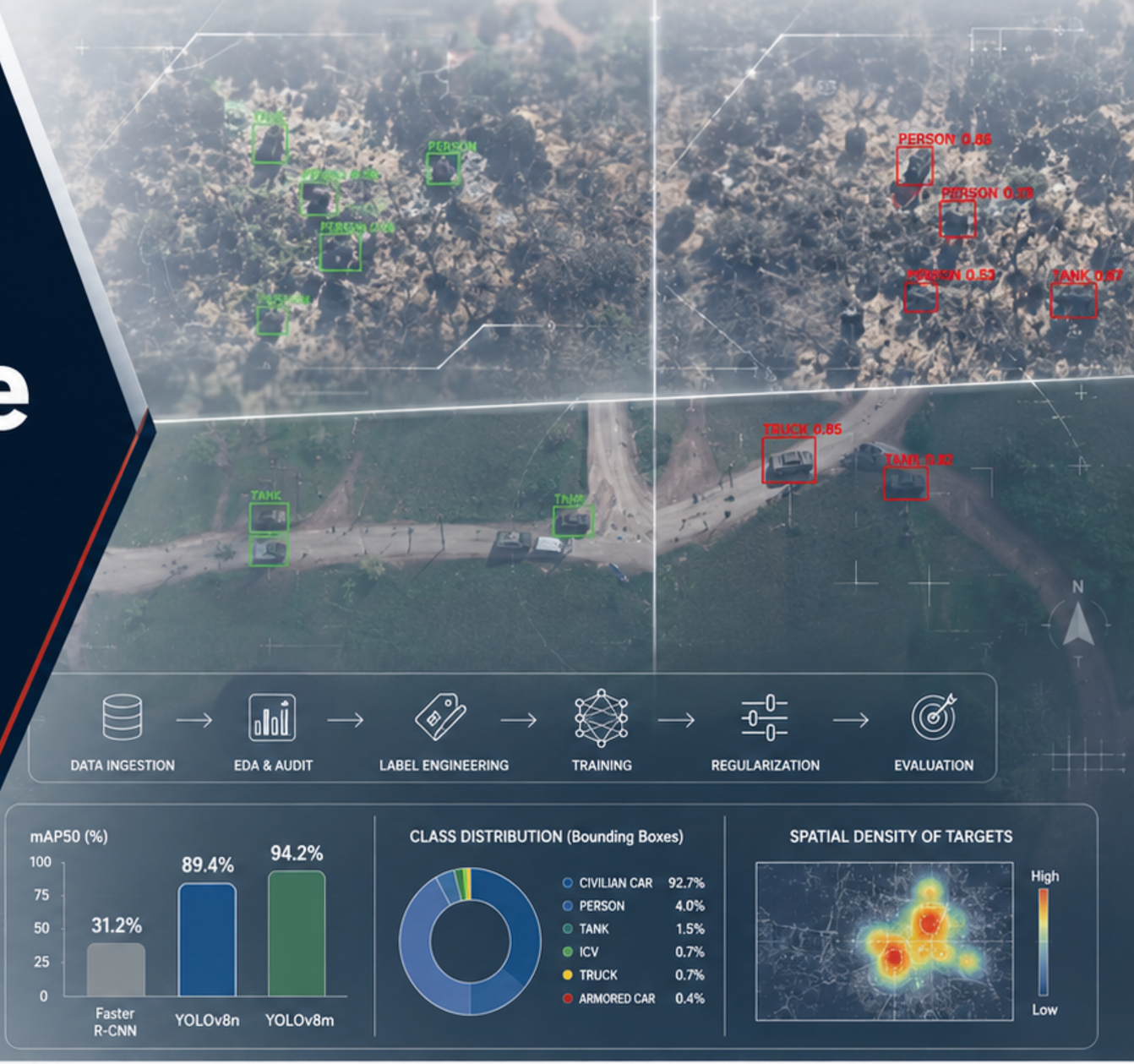
Military Object Detection Pipeline

Real-Time Detection of Tactical Assets under Civilian Noise and Domain Shift

Slide content in English · Oral presentation in Armenian

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Dataset engineering · detector design · domain generalization · deployment readiness



Dataset Engineering & Negative Mining

Why raw data curation was the critical first step

- EDA revealed severe class imbalance: civilian cars dominated the annotation space.
- Unfiltered training would bias the detector toward generic traffic patterns.
- We removed civilian-car emphasis and treated them as negative space during training.
- This forced the network to focus on tactical geometry: tracks, armor, turrets, and military silhouettes.

Civilian car annotations

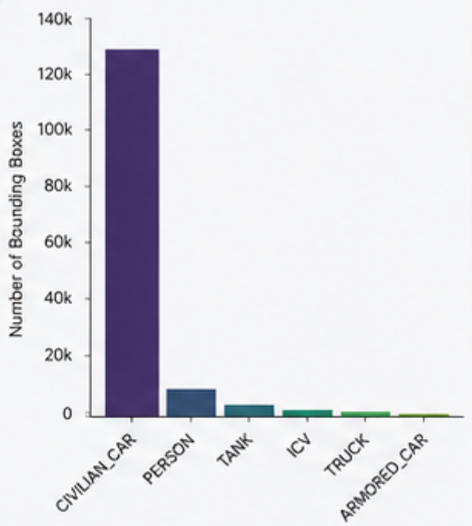
≈ 92.7%

of total bounding boxes

$$D^* = D \setminus \{\text{civilian-car dominant bias}\}$$

$$\min_{\theta} \mathbb{E}_{(x,y)\sim D^*} [L(f_\theta(x), y)]$$

Class distribution



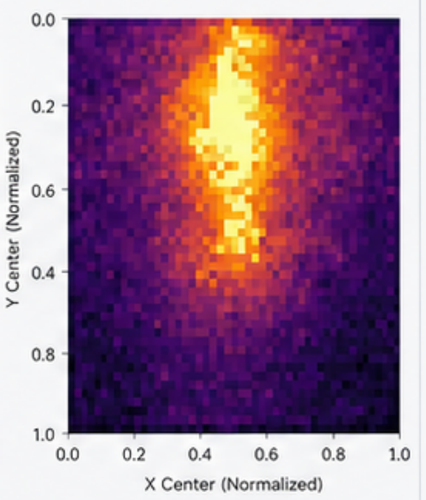
Civilian cars dominate the dataset.

Box geometry



Most boxes are small; military assets show broader geometry.

Spatial density



Targets concentrate near the center of the frame.

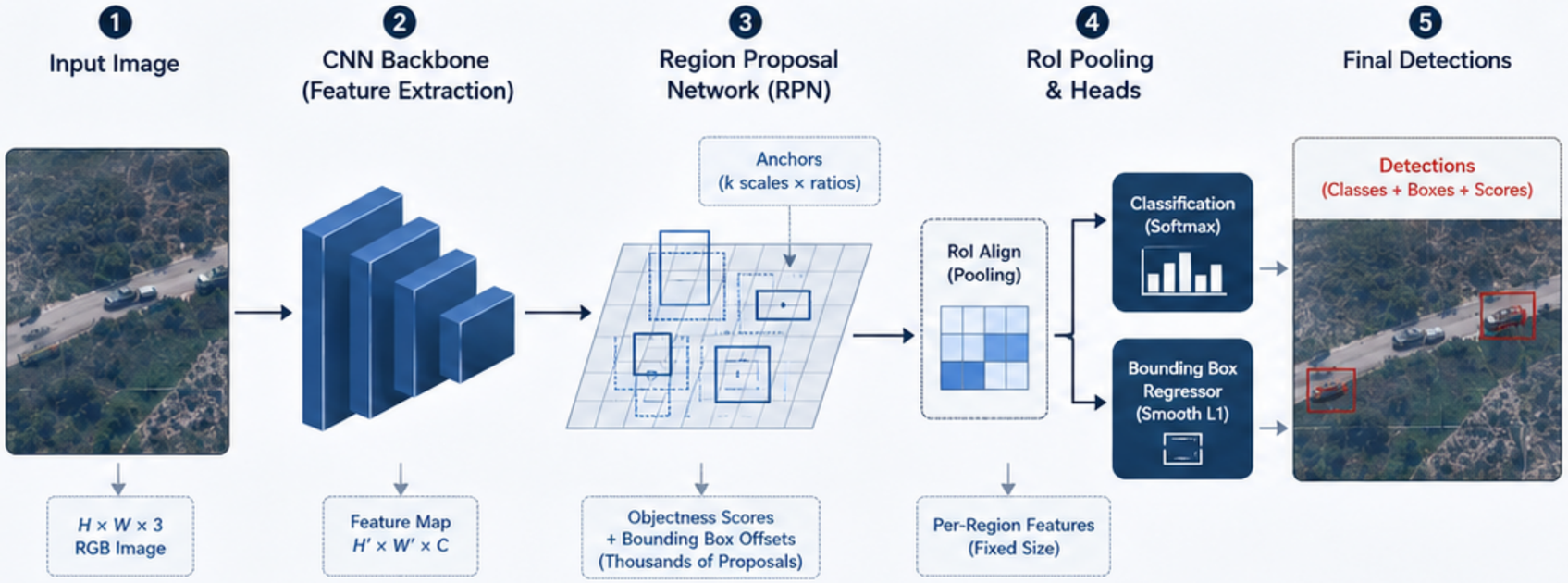
Sample frames



Diverse operating environments and perspectives.

The Architectural Bottleneck: Faster R-CNN

High theoretical precision, but operationally too slow for edge deployment



$$T_{total} = T_{backbone} + T_{RPN} + T_{RoI} + T_{head}$$

Two-stage pipeline ⇒ multiple heavy components ⇒ high end-to-end latency

Faster R-CNN: Characteristics & Trade-Offs

Detection paradigm	**Two-stage** (RPN + Detection Heads)
Latency	**High** (~150–300 ms / frame*)
Precision	**High** (State-of-the-art baseline)
Edge readiness	**Low** (Computationally heavy)
Suitability for 60 FPS defense workflows	**No** (Not real-time)

* Indicative range on embedded GPU-class hardware

- Thousands of overlapping region proposals create substantial computational overhead.
- Two-stage inference improves precision but increases end-to-end latency.
- For drones, autonomous turrets, and edge devices, SWaP-C constraints dominate architecture choice.
- Conclusion: unsuitable as the final production detector despite its strong academic baseline.

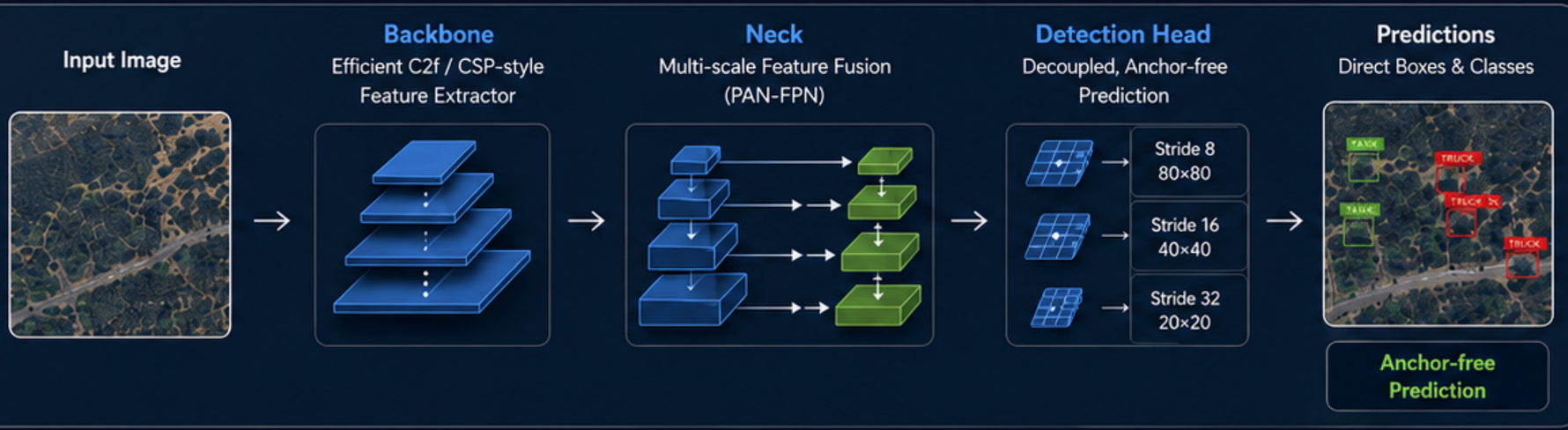
Requirement: real-time response >= 60 FPS

Decision: not selected for edge deployment

Lesson learned: precision alone is insufficient without deployment-grade speed.

The Pivot to YOLOv8

A single-stage detector aligned with real-time tactical deployment



1. YOLOv8 performs detection in a single forward pass, removing proposal-stage latency.
2. Anchor-free prediction improves flexibility across varied target scales and viewpoints.
3. Distribution Focal Loss and box-regression losses improve localization quality.
4. The architecture satisfies the project's real-time inference objective on modern GPUs.

Single-stage
One forward pass from image to detections.

Anchor-free
Greater adaptability across scales and aspect ratios.

>100 FPS target
Designed for real-time tactical deployment.

Training Objective (YOLOv8)

$$L = \bar{\lambda}_{box} L_{box} + \bar{\lambda}_{cls} \cdot L_{cls} + \bar{\lambda}_{dfl} L_{dfl}$$

L_{box} Box regression loss L_{cls} Classification loss L_{dfl} Distribution Focal Loss

Faster R-CNN
Two-stage detector
Higher latency

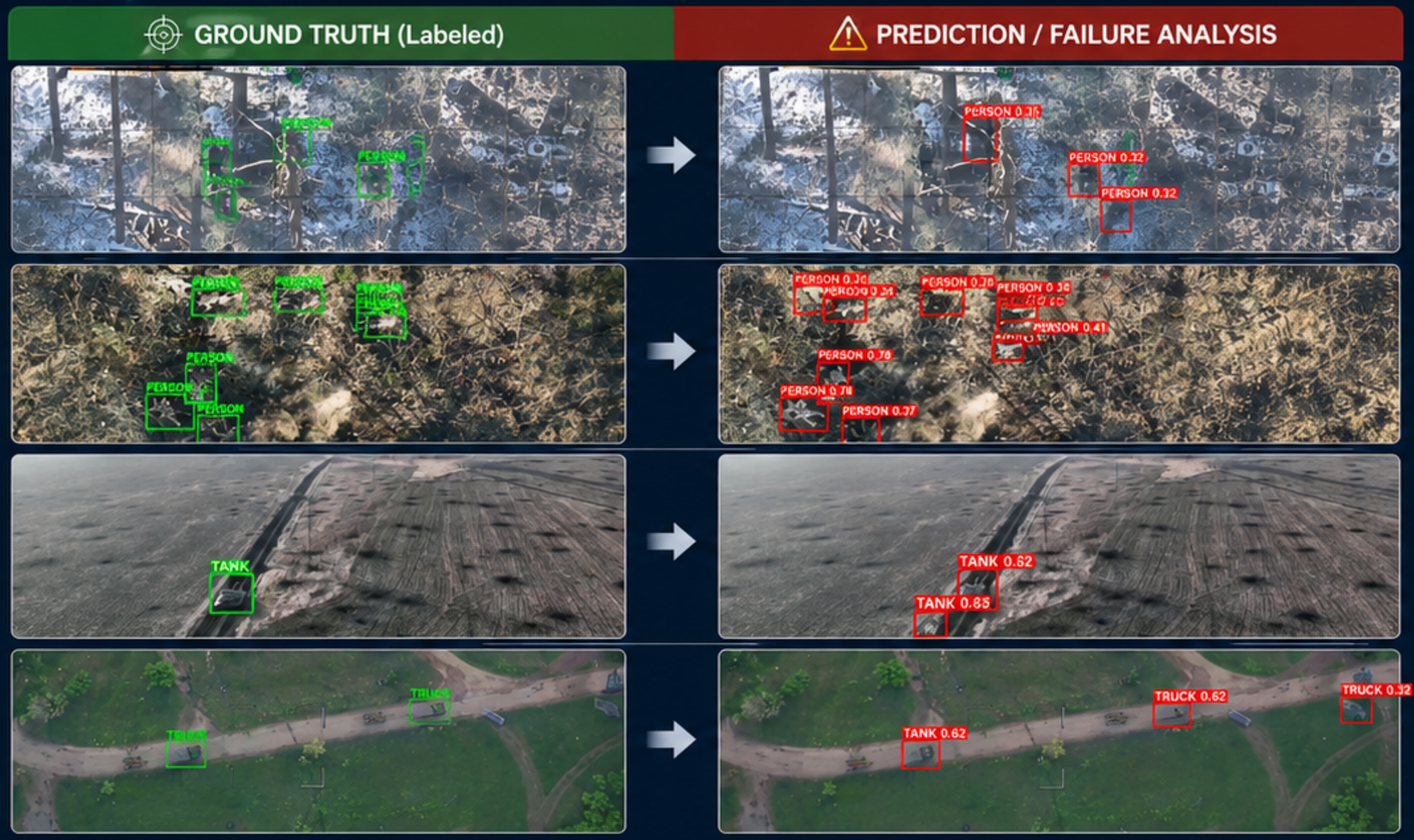
YOLOv8n
Single-stage detector
Major latency reduction

YOLOv8m
Stronger accuracy
Deployment-ready

YOLOv8 unlocked the speed-accuracy regime required for the application.

The Real-World Illusion: Domain Shift

Strong validation metrics did not guarantee field robustness



01 Around epoch 50, the baseline model reached **~89%** validation accuracy.

02 In unconstrained real footage, performance **degraded sharply.**

03 The detector had partially **overfit** to background context, lighting, and terrain statistics.

04 This is a classic **domain-shift / out-of-distribution** generalization failure.

From Seen to Unseen: The Domain Gap



Validation success ≠ deployment robustness

$$P_{train}(x) \neq P_{deploy}(x)$$

$$Risk_{deploy} > Risk_{val}$$ under distribution shift

We needed to generalize beyond texture and context memorization.

Solving Domain Shift: Aggressive Generalization

Capacity scaling plus augmentation-driven robustness

AUGMENTATION MATRIX

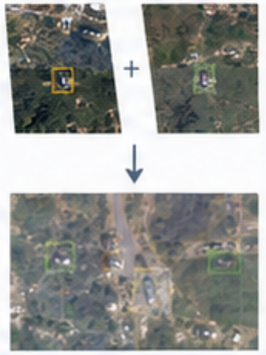
1. Mosaic

Stitches 4+ images into a single scene



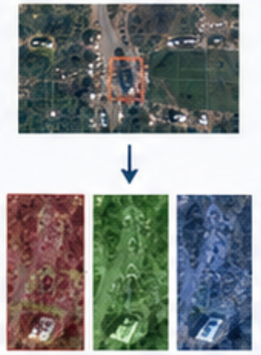
2. MixUp

Blends images and labels in pixel space



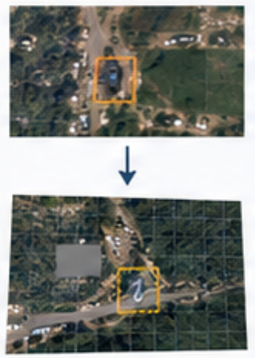
3. HSV Perturbation

Random shifts in Hue, Saturation, Value



4. Perspective Transform

Random perspective, scale and rotation



5. Random Erasing / Camouflage Simulation

Occludes regions and mimics camouflage



Model upgrade: YOLOv8m

higher representational capacity for complex tactical features



1. We increased model capacity to capture harder target signatures.
2. Heavy augmentation broke background shortcuts and encouraged shape-based reasoning.
3. Color, geometry, and occlusion perturbations approximated real deployment variability.
4. The training objective shifted from interpolation to robust invariance learning.

TRAINING OBJECTIVE

$$\min_{\theta} \; E_{(x,y)\sim\mathcal{D}} \; E_{T\sim\mathcal{A}} \left[\mathcal{L}\left(f_\theta\left(T(x)\right), y\right) \right]$$

A = augmentation distribution

Context Memorization (Shortcuts)



Model relies on spurious background cues.

Robust Target Representation (Invariance)



Model focuses on target shape and discriminative features.

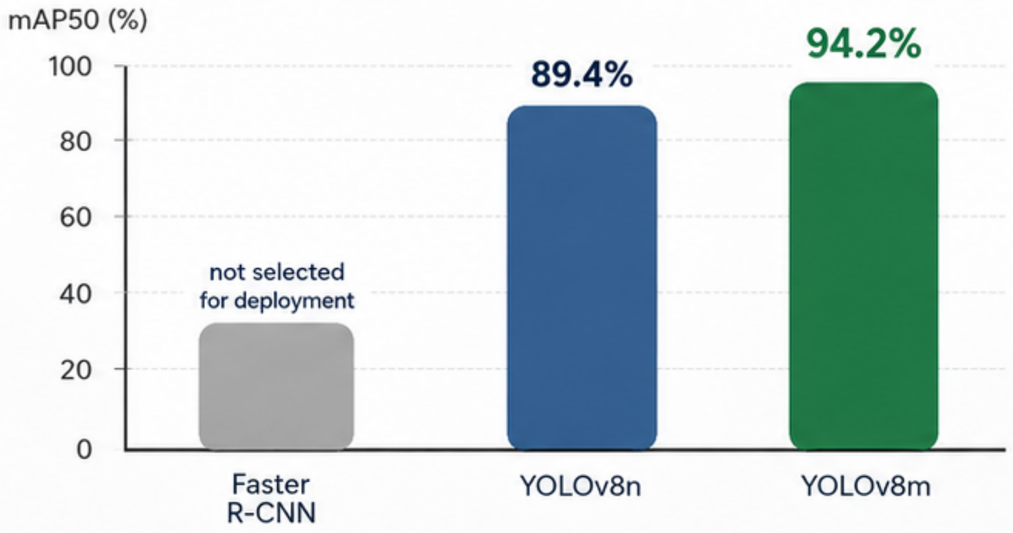
Generalization was engineered, not left to chance.

Final Results & Conclusion

From brittle prototype to deployment-ready tactical detector



Model Performance Progression (mAP50 @ IoU=0.50)



mAP50 measures area under precision-recall curve at IoU=0.50.

Key Takeaways

mAP50: 94.2%

High detection accuracy achieved with YOLOv8m at IoU=0.50.

Real-time inference readiness

Meets deployment criterion of real-time FPS with accuracy.

Lower civilian-noise susceptibility

Reduced false positives from civilian patterns and background clutter.

Improved robustness to lighting, occlusion, and viewpoint shift

Stronger generalization across domain shifts and environments.

Conclusions

- Data curation, not just model choice, was central to success.
- YOLOv8m provided the best balance of accuracy and deployment practicality.
- The final pipeline is suitable for continued field validation and future productionization.

End-to-End Summary

Dataset Ingestion — Curated datasets with normalized annotations →

EDA & Audit — Analyze distribution, class balance, and quality →

Label Engineering — Refine labels, remove civilian cars, hard negative mining →

Training Strategy — Faster R-CNN baseline → YOLOv8n pivot → YOLOv8m →

Regularization — Mosaic, MixUp, HSV, perspective, erasing →

Evaluation — mAP50, F1, FPS, qualitative error review

Thank you — Questions?

Production-ready intelligence begins with robust data and robust generalization.